RUHR UNIVERSITÄT BOCHUM RUB

Study Project (Bachelor/Master)

JUnit meets APR: Development of a JUnit Extension for Automated Program Repair

Motivation and Background: Automated program repair (APR) [CACM'2019] promises great help for software developers by automatically generating patches. Various techniques for APR (search-based, semantic-based, template-based, ML-based) have been proposed in the research community, but only a few have made their way to the application in practice. This project aims to provide a concrete solution that integrates with existing workflows like JUnit. JUnit is the standard unit testing framework for Java. Failing test cases usually indicate a regression error, meaning that an error was introduced with recent changes. An automated repair solution for JUnit would follow up on the failing test case event and automatically propose a patch for the developer. Therefore, it attempts to merge testing and repair, which is an essential step to improve software quality at scale.

Student Task and Responsibilities:

- IDE (VSCode or IntelliJ) extension that allows the developer to run tests with JUnit followed by the automated repair of the failing tests by adapting the corresponding Java code. This represents the complete workflow from bug detection, fault localization, and fix generation and application.
- The students are supposed to explore existing APR techniques and should also implement their own variant/new technique. A starting point can be existing APR libraries [ISSTA'2016].
- The implementation(s) must be evaluated on open-source projects, which can be selected by the students. The findings of the conducted experiments must be thoroughly documented.

Deliverables: Source code and its documentation, and a publishable report (incl. the evaluation results), ideally to submit to a conference or at least publish as a technical report on arxiv.org.

Number of Participants: 2-4

Industrial partner: None (done at RUB)

Prerequisites: Programming experience, preferably in Java, is needed for this project. Further, it would be beneficial to have experience with JUnit Testing and IDE plugin development. Prior knowledge in automated program repair is not needed.

[CACM'2019] Claire Le Goues, Michael Pradel, and Abhik Roychoudhury. 2019. Automated program repair. Commun. ACM 62, 12 (December 2019), 56–65. https://doi.org/10.1145/3318162

[ISSTA'2016] Matias Martinez and Martin Monperrus. 2016. ASTOR: a program repair library for Java (demo). In Proceedings of the 25th International Symposium on Software Testing and Analysis (ISSTA 2016). Association for Computing Machinery, New York, NY, USA, 441–444. https://doi.org/10.1145/2931037.2948705

[JUnit] https://junit.org/junit5/

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